Possession Without the Ball: Frame-Level Possession from Team Shape, Transferred Labels and a Three-State Decoder

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More than a Hack 2027 · Phase 1: Ball Possession

`https://github.com/Feras005/mtah2027-dsba`

Abstract

We label every frame of a professional football match as home possession, away possession or dead ball, using automatic and uncorrected tracking in which the ball is unobserved on 41% of graded frames. We read possession from the players as well as the ball. Two gradient-boosted models trained on *real labels from three other matches* (Metrica Sports' public sample data) answer the two questions of a three-state hidden Markov model: *is the ball in play?*, from restart formations and crowd dynamics, and *which team has it?*, from team shape and the ball, trained with the ball hidden at random. A third, ball-free shape model is trained on rule-generated pseudo-labels from the match itself. Viterbi decoding charges a high price for a change of team, and hand-overs are then snapped to the touch that caused them. Where the tactical tracker loses the ball, the broadcast feed, aligned to it by player positions alone, fills in. Every setting was chosen by leave-one-game-out SPELLF1 on the external matches and checked on the competition match with a ground-truth-free self-consistency test. The learned models and snapping raise external SPELLF1 from 0.540 to 0.581, while self-consistency rises from 0.825 to 0.854. No competition ground truth was used to choose anything, and the pipeline is deterministic.

1 Introduction

Possession is the first thing an analyst segments in a match and the first thing an automatic system gets wrong at the margins: restarts, contested balls and deflections. Phase 1 of More than a Hack 2027 [13] asks for a possession label on each of the 199,319 frames of the tactical video of FC Barcelona vs Athletic Club (LaLiga 2024–25), ranked on SPELLF1, which scores each possession as a whole object rather than as a sequence of frames.

Two properties make the problem hard. First, the tracking is automatic and uncorrected, with no player identities, and the ball is missing on 41% of graded frames, disproportionately in crowded boxes, aerial duels and tackles, which are exactly the moments at which possession changes. Second, SPELLF1 punishes fragmentation and boundary error severely: a per-frame classifier that flickers can reach a respectable MACROF1 while reporting several times more possessions than the match contained.

We treat the task as *segmentation under heavy occlusion*, with the following contributions:

- **Transferred supervision.** Event data from three public matches are converted to possession

labels by a fixed rule and used to train a ball-free dead-ball model and a team model that is trained with the ball hidden, giving real labels without hand-labelling the competition match.
- **A ball-free possession signal.** Collective team shape, oriented by an attacking direction inferred from the data, predicts possession where the ball is absent.
- **A decoder built around the ranking metric.** A three-state HMM separates *whether the ball is alive* from *which team has it*; its switch cost formalises controlled possession, and boundaries are snapped to touches.
- **A second camera without calibration.** The broadcast feed is aligned to the tactical timeline from player positions alone and used to fill ball gaps.
- **Ground-truth-free model selection.** Leave-one-game-out SPELLF1 on external matches chooses settings; a self-consistency SPELLF1 checks them on the competition match.

2 Related Work

Possession from tracking data. Link and Hoernig [10] define individual ball possession from player and ball positions and dynamics, and Vidal-Codina et al. [17] detect possession gains, losses and set-piece triggers from tracking, noting that the absence of an in-play flag is a key limitation. Both assume a largely continuous ball trajectory; our setting removes it on 41% of frames, which motivates evidence that does not depend on the ball.

Team shape and space. Lucey et al. [11] and Bialkowski et al. [1] represent team strategy and formation from spatiotemporal data, and Fernández and Bornn [4] quantify space creation. Possession-value models [5, 16] take possession as given; we infer it, using the same intuition that the team in possession expands the pitch.

Sequence decoding, weak supervision and transfer. We decode with a hidden Markov model and the Viterbi algorithm [15, 18]. All learned components are gradient-boosted trees [6, 7] as implemented in scikit-learn [14]. The in-match shape model uses rule-generated pseudo-labels in the spirit of Lee [8], with features ranked by permutation importance [2]; the transferred models use Metrica Sports' public sample data [12]. The crowd-speed model is a Gaussian mixture fitted by EM [3]. The ranking metric averages matching F1 over IoU thresholds, the COCO convention [9] applied to time.

3 Problem Setting and Data

Task. For each video frame t, output $y_t \in \{\mathtt{H}, \mathtt{A}, \mathtt{D}, \mathtt{X}\}$: home or away in possession with the ball alive, dead ball, or abstention. Only the two halves are graded (158,857 frames). A *possession* is a maximal run of `H` or of `A`; stoppages are not possessions.

Metrics. Let $\mathscr{G}$ and $\mathscr{P}$ be the true and predicted possessions. At threshold τ, same-team pairs are matched greedily one-to-one in descending temporal IoU, keeping pairs with IoU $\geq \tau$; with TP_τ matches,

$$F_\tau = \frac{2\,\mathrm{TP}_\tau}{|\mathscr{P}| + |\mathscr{G}|}, \qquad \text{SPELLF1} = \tfrac{1}{5} \sum_{\tau \in \{0.3,0.4,0.5,0.6,0.7\}} F_\tau. \tag{1}$$

MACROF1, the mean per-class F1 over `H`, `A`, `D`, breaks ties. Before SPELLF1 is computed, each `X` is forward-filled with the previous label in the half.

Assets. Table 1 lists what was used.

Training data. Table 2 lists every labelled dataset. No hand labels on the competition match, no competition ground truth (public or private) and no pre-trained models were used. The unsupervised components are fitted on the competition match's tracking without labels.

Table 1: Assets used and dropped.

Asset	Use
Tactical ATD	Ball, proximity, team shape, restart formations, crowd speed.
Smart Tagging	Set pieces as restart anchors; phases as a weak team prior with learned direction; kickoff side.
Broadcast ATD	**Ball only**: aligned to the tactical timeline by player positions; fills ball gaps.
Homography, videos	**Not used**: both ATDs are already in pitch coordinates.
Public 5-minute ground truth	**Only** to verify frame alignment and file format. No setting was chosen with it.
Metrica Sports sample data [12]	External training and validation data (Table 2).

Table 2: Training data.

Dataset	Source	Labelled by	Size
Metrica sample games 1–3	Metrica sample-data repository [12], commit `e706dd5`; public, anonymised; used with acknowledgement as its README asks	Metrica Sports' event data, converted to `H`/`A`/`D` by a fixed rule (`possession/external.py`)	3 matches; 427,894 labelled frames (34% dead); 1,044 possessions
Pseudo-labels on BAR-ATH	Tactical ATD and Smart Tagging of the competition match	A fixed rule (`possession/pseudo.py`); no person	15,283 team frames; 16,759 dead frames

Labels from events. The team in possession is the team of the latest on-ball event (pass, carry, shot, recovery, ball lost, set piece), so possession changes at the first on-ball event of the other side, the interception a `RECOVERY` marks. The ball is dead from a `BALL OUT`, a shot that goes out or in, a foul or an offside, until the restart that follows (a `SET PIECE` or a goal-kick pass).

4 Method

4.1 Ball evidence

Ball detections implying speeds above 40 m/s are discarded and occlusions of at most 0.4 s are bridged. With d_t^H, d_t^A the distances in metres from the ball to the nearest home and away player, the proximity log-odds are

$$e_t = \gamma_t \operatorname{clip}\left(d_t^A - d_t^H, -3, 3\right), \tag{2}$$

where $\gamma_t \in [0,1]$ is reduced when no player is within about 2.5 m and when the ball is in flight (above 7 m/s). A loose ball thus carries no evidence and possession stays with the last team. The ball is out of bounds when observed more than 0.75 m outside the lines.

4.2 Broadcast ball

The broadcast ATD runs on its own timeline, with cuts and replays, and no mapping is provided. Both feeds record the same match in real time, so live broadcast footage sits at a fixed frame offset from the tactical video. We find it from the players alone: for 400 sampled broadcast frames, the cost of each candidate tactical frame is the median distance from each broadcast player to the nearest tactical player, one-way because the camera sees a subset. 391 of 400 frames matched and voted for exactly two offsets, one per half (4,014 and 26,914 frames). A broadcast frame is then accepted only where its players match within 1 m, which rejects replays and close-ups. The feeds align on 57% of graded frames and, where both see the ball, agree to 1.2 m (median).

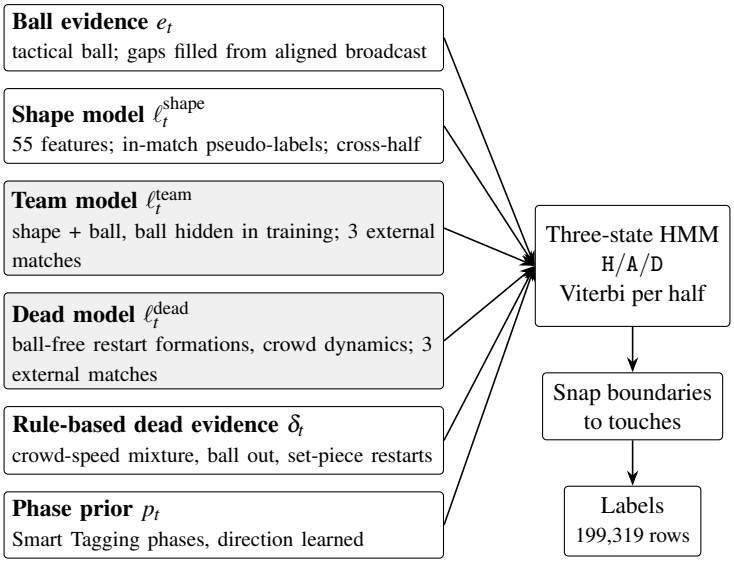


Figure 1: Overview. Shaded components are trained on external matches. Every component emits per-frame log-odds: e_t, ℓ_t^{shape}, ℓ_t^{team} and p_t answer *which team*; ℓ_t^{dead} and δ_t answer *alive or dead*.

4.3 Models trained on other matches

Both models are `HistGradientBoostingClassifiers`, one per decoder question.

Dead model: P(ball out of play), ball-free by design. In the sample matches the ball is almost never tracked while play is stopped (7–21% of dead frames, against about 65% in BAR-ATH). A model that saw the ball would learn "no ball means stoppage", an artefact that would not transfer. It therefore learns stoppages from *positive evidence in the players*: restart formations (the nearest player to a touchline and players beyond the lines for throw-ins; the nearest player to a corner flag and players in each box for corners; players in the six-yard boxes and the nearest player to a goal line for goal kicks; players in the centre circle and mean distance from the halfway line for kick-offs), crowd dynamics (crowd speed now and over 3, 6 and 12 s, the 6 s minimum, the 3 s before and after, players standing still), and the 55 team-shape features of §4.4.

Team model: P(home | alive). Team shape plus ball features (proximity, distances, ball speed, whether the ball is known). In training, the ball is hidden on random 1–8 s blocks covering 40% of frames, so the model learns to fall back on shape when the ball is gone.

For BAR-ATH both models are trained on all three matches. Their log-odds enter the decoder with weights 0.2 (dead) and 0.5 (team), chosen in §5.3.

4.4 In-match shape model and pseudo-labels

For each team we compute centroid, width, length, spread, height of the deepest outfield player, most advanced player, players in the opponent half, speed, *push* (centroid velocity towards goal) and free space (mean distance to the nearest opponent), together with home-minus-away differences, their 1 s means and 2 s trends: 55 ball-free features, measured towards each team's attacking goal, whose direction is inferred per half from the data. The model is trained *cross-half* on pseudo-labels from the competition match, so every prediction is out of sample. *Team* pseudo-labels: the ball is observed and slower than 5 m/s, a player is within 1 m of it and no opponent within 3 m. *Dead*: the ball is observed out of bounds, a tagged set piece precedes its estimated restart, or the frame precedes the published second-half kickoff. Its log-odds enter with weight $0.2\,w_t$, where the gate w_t is 0 when a player is clearly on the ball.

4.5 Rule-based dead evidence and phase prior

δ_t sums: ball observed out of bounds ($+4$); a two-component Gaussian mixture fitted without labels on log crowd speed (components at 0.66 and 1.54 m/s); ball rolling inside the pitch (-0.7) or unseen ($+0.15$); and set pieces, dead before the estimated restart and live after it. Set-piece tags are fixed 35 s windows in which play resumes only 20–30% of the way in, so the restart is estimated per set piece as the largest rise in crowd speed within the window.

Each Smart Tagging phase code contributes at most ± 0.25 log-odds towards its tagged team, with the *sign learned* from frames where proximity is confident: every `DEFENDING ...` code came out negative (the tag names the team *without* the ball), `BUILD UP` and `PROGRESSION` positive.

4.6 Decoder and snapping

States are decoded per half by Viterbi,

$$\hat{y}_{1:T} = \arg\max_{y_{1:T}} \sum_{t=1}^{T} \Big[\phi_t(y_t) - c_t(y_{t-1}, y_t) \Big], \tag{3}$$

where the log-emissions ϕ_t use team evidence only to choose between `H` and `A`, never to make a live state more likely than `D`; that is left to the dead evidence. Transition costs are given in Table 3. Since $|e_t| \leq 3$, a change of team on proximity alone requires at least $\lceil 14/3 \rceil = 5$ frames of saturated evidence: the implemented notion of *controlled* possession. Possessions under 0.8 s and stoppages under 1.2 s are absorbed. Half 1 starts live, biased towards the tagged kickoff side; half 2 is `D` until the published kickoff (frame 113,742).

Table 3: Transition costs in log-likelihood units.

Transition	Cost	Reduced where
`H` $\leftrightarrow$ `A`	14	Never: a touch, deflection or lost aerial duel cannot pay for it.
live $\rightarrow$ dead	7	To 1.5 near an observed ball out of bounds.
dead $\rightarrow$ live	7	To 2 within 3 s of an estimated restart.

Snapping. After decoding, each hand-over is moved to the first touch by the new team after the old team's last touch, each restart to the first touch by the restarting team, and each stoppage to the first frame the ball is seen out, if one lies within 1 s. A touch is a jump in ball velocity of at least 4 m/s with a player within 2 m. This follows the event-label convention, in which possession changes at the interception.

Possession rules as implemented. A loose or contested ball produces no team evidence, so the last team keeps possession. A change of possession requires sustained control and is then placed at the new team's first touch. At restarts, the set piece anchors the moment play resumes, and the side comes from the tag where it names one, otherwise from proximity (throw-ins are tagged `N/A`). A dead ball is always `D`.

5 Experiments

5.1 Validation protocols

Leave-one-game-out SPELLF1 on external matches (selection). The full pipeline runs on each sample match exactly as on BAR-ATH, except for Smart Tagging, which they do not have, with the learned models trained on the other two matches. Output is scored by the *official scorer* against the event labels. This is the criterion for every setting.

Self-consistency SPELLF1 on BAR-ATH (check). The match is decoded with the ball as tracked, then again with the ball hidden on random 1–8 s blocks covering 40% of frames, and the second

decode is scored against the first with the official scorer, over three seeds. It scores whole possessions, so a possession split by a false stoppage shows directly. On its own it cannot select settings, because a decoder that never changes its mind agrees with itself perfectly, so it is a check rather than a target.

Hypotheses. **H1**: labels transferred from other matches improve segmentation. **H2**: snapping boundaries to touches improves SPELLF1 beyond the decoder alone. **H3**: the gains are not bought by switching less. **H4**: the aligned broadcast restores possessions lost to ball occlusion.

5.2 Transferred models

Table 4 reports leave-one-game-out discrimination. Both models transfer across matches, and the team model stays above 0.92 AUC although the ball is hidden on 40% of its training frames. The in-match shape model reaches AUC 0.934 and 0.940 on the held-out halves against pseudo-labels (logistic regression 0.906 and 0.913); its most important features say that *the team with the ball is the wider one*.

Table 4: Leave-one-game-out AUC of the transferred models.

Held-out match	Dead model	Team model
Sample game 1	0.832	0.921
Sample game 2	0.869	0.945
Sample game 3	0.897	0.942

5.3 Ablations on external matches

Table 5 and Figure 2 report leave-one-game-out SPELLF1. Each learned model helps alone and more in combination, and snapping adds a further 0.010, supporting **H1** and **H2**. Self-training was tried and rejected (§5.5).

Table 5: Leave-one-game-out SPELLF1 on the external matches (true possession count 348).

Configuration	Mean SPELLF1	Per match	Possessions
v2 pipeline, as first submitted	0.540	0.540 / 0.550 / 0.528	260
+ dead model, weight 0.2	0.563	0.560 / 0.576 / 0.553	261
+ team model, weight 0.5	0.560	0.552 / 0.563 / 0.564	276
+ both	0.571	0.554 / 0.577 / 0.583	273
+ snapping to touches (v3, submitted)	**0.581**	0.556 / 0.588 / 0.601	270
+ self-training, 1 round	0.565	0.556 / 0.563 / 0.577	265
+ self-training, 2 rounds	0.566	0.559 / 0.566 / 0.574	263

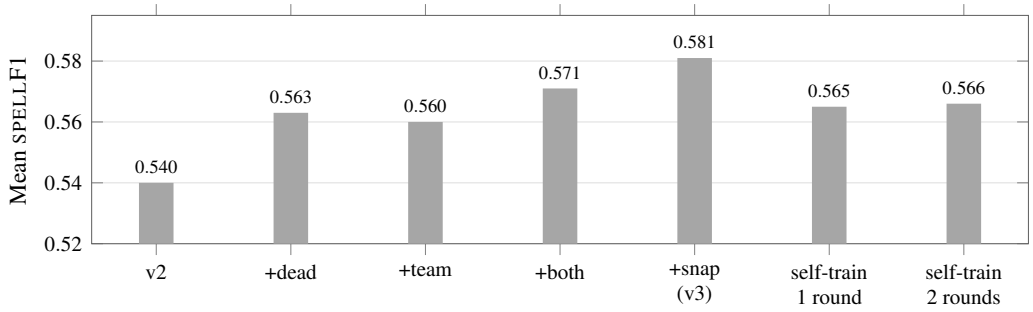


Figure 2: Leave-one-game-out SPELLF1 by configuration. Self-training lowers the score.

Decoder settings. The transition costs, originally set by hand as football priors, were tuned by coordinate ascent over a grid (two rounds): switch cost 5–18, stop and restart costs 3–12, minimum stoppage 0.4–3 s, minimum possession 0.2–1.2 s, and model weights. The hand-set values were already the best: switch cost 14 beat 5 (0.558), 9 (0.560), 11 (0.565) and 18 (0.571), and stop and restart cost 7 beat 3, 5, 9 and 12. Two changes scored marginally higher (minimum stoppage 2 s, +0.0024; shape weight 0.4, +0.003). We adopt only changes worth at least 0.005, since smaller ones cannot be distinguished from fitting three matches, so neither was taken.

5.4 Checks on the competition match

Table 6 reports self-consistency. Version 3 is more stable *and* finds more possessions, so the gain does not come from switching less (**H3**).

Table 6: Self-consistency SPELLF1 on BAR-ATH (mean of three seeds).

Version	Self-consistency SPELLF1	Possessions, ball visible
v2	0.825	301
+ learned models	0.851	325
+ snapping (v3)	**0.854**	323

Broadcast. The broadcast fills the ball on 3.6% of graded frames (59.3% → 63.0% observed), because both feeds tend to lose the ball at the same moments. To test it, the tactical ball is hidden on 40% of frames and refilled from the broadcast; against the decode with the ball visible, SPELLF1 rose on all three seeds (0.873 → 0.887, 0.838 → 0.874, 0.851 → 0.899), supporting **H4**. On the real match it changes 0.6% of labels.

5.5 Tried and dropped

- **Self-training** (decode, take frames at least 2 s inside decoded runs of at least 6 s as new labels, retrain, repeat): about 107,000 extra frames per match, but SPELLF1 fell (0.571 → 0.565 / 0.566). The model learns its own mistakes back.
- **A dead-ball shape model on pseudo-labels** lowered the version 2 frame-level masking score at every weight; the dead model trained on real labels replaces it.
- **Writing `X` mid-possession** (`--abstain`): `X` is forward-filled, so SPELLF1 cannot change, while MACROF1 drops.
- **Where the public ground truth places a switch** was not studied; the touch convention follows from the event labels of the external matches.

Version 2's settings (shape weight 0.2, team-only emissions, restart estimate) were chosen earlier by a frame-level ball-masking test (balanced recovery 0.699 → 0.780). It scores frames rather than possessions, which is why the two protocols above were added.

5.6 Final output

The submission contains 327 possessions (mean 12.1 s, median 7.5 s) and 121 stoppages, and 38% of graded frames are `D`. As a correctness check only, on the public five minutes it scores SPELLF1 0.518 (version 2: 0.422) against 0.326 for the published floor `baseline_smooth50`, and MACROF1 0.80 (version 2: 0.68). Seventeen possessions cannot rank anything, and nothing was chosen with this window.

6 Discussion and Limitations

- **Labels from another provider's events** are not the competition's ground truth: where a switch is placed, and how a stoppage is cut, may differ. More matches (the PFF FC World Cup 2022 data, 64 games, which require registration and were not used) would also reduce the noise on every

comparison above.

- **Possessions are still under-counted** on the external matches (270 against 348). The missing ones are short, and lowering the switch cost to catch them costs more than it gains.
- **The dead model sees no ball** by design. It cannot use a ball last seen near a line, because in the training matches the ball is not tracked during stoppages.
- **The broadcast adds little ball** (3.6% of frames), because both feeds tend to lose it at the same moments.
- **Smart Tagging cannot be validated externally**, since the sample matches have none.

7 Conclusion

When the ball is unobserved on two frames in five, possession can still be read from how the two teams occupy the pitch. Labels transferred from other matches, a ball-free dead-ball model built on restart formations, a decoder whose costs reflect how rarely possession truly changes, and boundaries snapped to touches raise external SPELLF1 from 0.540 to 0.581 and make the segmentation more stable under occlusion, with every choice made without the competition's ground truth. The main open problem is placing short possessions and their boundaries, where events from a different provider are only an approximation of the target.

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A Reproduction

```
# Python 3.12. Linux/macOS: python3 -m venv .venv, then .venv/bin/python
py -m venv .venv
.venv/Scripts/python -m pip install -r requirements-solution.txt
# unpack the data bundle into ./data (data/tactical/..., data/broadcast/...)
# Metrica sample data at a pinned commit (needs git):
.venv/Scripts/python fetch_external.py
# writes and validates evaluation/results/DSBA.csv:
.venv/Scripts/python run_pipeline.py
```

One command after the two downloads, about 6.5 minutes on a laptop the first time; parsing, the broadcast alignment and the feature matrices are then cached under `data/**/.cache/` (gitignored). The pipeline is deterministic: two runs, and a fresh clone with the pinned requirements, give the identical file with MD5

`2d6c737b376dc0fd110fd681b913adb2`

`python run_external.py study|tune|extras` re-runs the external studies, `python run_consistency.py` the self-consistency test and `python run_broadcast.py` the alignment. The flags `--no-external`, `--no-broadcast` and `--no-snap` switch each version 3 component off; with all three the pipeline is version 2.

B Synthetic data and data handling

`synthetic/make_synthetic.py` simulates a full match in the bundle's exact file formats, with its own ground truth. It was used to build and debug the pipeline before the gated data arrived, because the competition terms forbid sharing match data with cloud-hosted tools. It is not a model of the real match and chose no setting. `data/` (competition bundle, external sample data, caches) and `synthetic_data/` are gitignored, and the pipeline prints only counts, shares, fitted aggregates and model metrics, never tracking rows.